

Wasserstein Space and Otto Calculus

Optimal Transport and Stochastic Analysis · Lecture 3 of 3

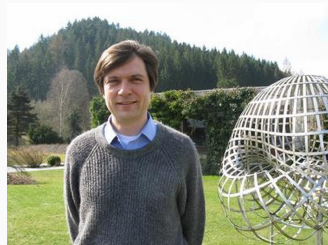
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Where we are

- ▶ **Lecture 1.** Brenier maps $Y = \nabla\varphi(X)$; McCann interpolation $X_t = (1 - t)X + t\nabla\varphi(X)$; Benamou–Brenier and the continuity equation.
- ▶ **Lecture 2.** Entropy regularization, Schrödinger bridges, Sinkhorn. Corrections like $\frac{\varepsilon}{2}\nabla\log\mu$ and the Fisher information $I(\rho)$ appeared.
- ▶ **Today.** A calculus on probability measures: velocities, gradients, Hessians, gradient flows.
- ▶ Goal: you should be able to do these computations yourself. The rigorous theory is Ambrosio, Gigli & Savaré (2008).



Felix Otto

The idea in one slide

	Euclidean calculus	Otto calculus
point	$x \in \mathbb{R}^d$	$\rho = \text{Law}(X)$
curve	$t \mapsto x_t$	Curve of probabilities $t \mapsto \rho_t$
velocity	vector v	a vector field: $v \in L^2(\rho)$
squared speed	$\ v\ ^2$	$\mathbb{E} \ v\ _{L^2(\rho)}^2$
function	$f(x)$	$F(\rho)$, e.g. $\mathbb{E}_\rho V(X)$, $\text{Ent}(\rho)$

Slogan

Otto calculus is Euclidean calculus done **inside an expectation**. $\rho = \delta_x$ gives back $\mathbb{R}^d$.

Curves and velocities

How to draw a curve of probabilities?

- ▶ Curves on probabilities $\Leftrightarrow$ Velocity fields $v_t(x) : \mathbb{R}^d \rightarrow \mathbb{R}^d$.
- ▶ Particles move by an ODE: $\dot{X}_t = v_t(X_t)$, $X_0 \sim \rho_0$. Let $\rho_t = \text{Law}(X_t)$.
- ▶ $t \mapsto \rho_t$ gives a curve of probabilities.
- ▶ $t \mapsto X_t$ is the Lagrangian description.

Continuity equation revisited

- For a test function ζ , the chain rule inside $\mathbb{E}$:

$$\int \zeta(x) \dot{\rho}_t(x) dx = \frac{d}{dt} \mathbb{E} \zeta(X_t) = \mathbb{E} [\nabla \zeta(X_t) \cdot v_t(X_t)] = \int \nabla \zeta(x) \cdot v_t(x) \rho_t(x) dx.$$

- Written with densities, this is the **continuity equation** (Eulerian description):

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.$$

- Lagrangian (particles) and Eulerian (densities) are two descriptions of the same curve.

Example 1: Gaussian

- ▶ Example: $d = 1$, $v_t(x) \equiv 1$, $\rho_0 = N(0, 1)$.

$$\partial_t \rho_t(x) + \partial_x \rho_t(x) = 0.$$

- ▶ Solution $\rho_t = N(t, 1) = \frac{1}{\sqrt{2\pi}} e^{-(x-t)^2/2}$.
- ▶ How? Transport equation. Particles moving with unit velocity:

$$\dot{X}_t = 1.$$

- ▶ If $X_0 \sim N(0, 1)$, then $X_t \sim \rho_t$.

Example 2: scaling an exponential

- ▶ Let $\rho_t = \text{Exp}(1+t)$, $0 \leq t \leq 1$. $\rho_t(x) = (1+t)e^{-(1+t)x}$, $x \geq 0$.
- ▶ If $X \sim \text{Exp}(1)$, $a > 0$ then $X/a \sim \text{Exp}(a)$.

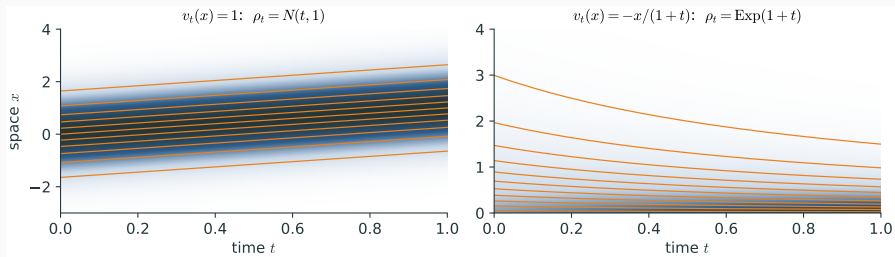
$$X_t := \frac{X_0}{1+t} \sim \rho_t.$$



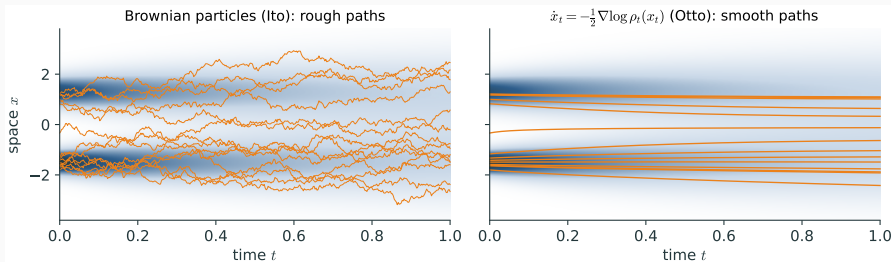
$$v_t = \dot{X}_t = -\frac{X_t}{1+t}, \quad v_t(x) = -\frac{x}{1+t}.$$

- ▶ Try $\rho_t \sim \text{Exp}(t)$, $0 < t \leq 1$.

Characteristic lines are smooth



Example 4: Brownian particles still have a velocity



- ▶ $X_t = X_0 + W_t$ has no derivative. But its law solves $\partial_t \rho_t = \frac{1}{2} \Delta \rho_t = \nabla \cdot (\rho_t \cdot \frac{1}{2} \nabla \log \rho_t)$.
- ▶ So $v_t = -\frac{1}{2} \nabla \log \rho_t$: right panel, particles following $\dot{x}_t = v_t(x_t)$ have the same marginals.

What kind of curves are given by this correspondence?

- ▶ Curves $(\rho_t, 0 \leq t \leq 1)$ is absolutely continuous (AC) if

$$\mathbb{W}_2(\rho_s, \rho_t) \leq \int_s^t g(u) du, \quad \forall 0 < s \leq t < 1,$$

Theorem AGS: AC curves $\Leftrightarrow$ Continuity equations

- ▶ Suppose $(\rho_t, 0 \leq t \leq 1)$ is AC. Then, for some $(v_t \in L^2(\rho_t))$,

$$\partial_t \rho_t + \nabla \cdot (v_t \rho_t) = 0.$$

- ▶ Conversely, if (ρ_t, v_t) solves cont. eqn. and $\int_0^1 \|v_t\|_{L^2(\rho_t)} < \infty$, then (ρ_t) is AC.

Example

- Let $(\theta_t, 0 \leq t \leq 1)$ be an AC Euclidean curve in $\mathbb{R}^d$. Let $\rho_t = \delta_{\theta_t}$. Then

$$\mathbb{W}_2(\rho_s, \rho_t) \leq \int_s^t \|\dot{\theta}_u\| du.$$

- Or, $\rho_t = N(\theta_t, 1)$. Even then,

$$\mathbb{W}_2(N(\theta_s, 1), N(\theta_t, 1)) = \|\theta_s - \theta_t\| \leq \int_s^t \|\dot{\theta}_u\| du.$$

- Example of a **non AC curve** $\rho_t = \text{Bern}(t)$, $0 \leq t \leq 1$.

$$\mathbb{W}_2(\text{Bern}(s), \text{Bern}(t)) = \sqrt{t - s}.$$

The same curve is given by many stochastic processes

- ▶ Consider a curve (ρ_t, v_t) , $t \geq 0$. One process that has this marginal flow: $X_0 \sim \rho_0$,

$$\dot{X}_t = v_t(X_t).$$

- ▶ Here's an entire family of processes tracing the same marginal flow.

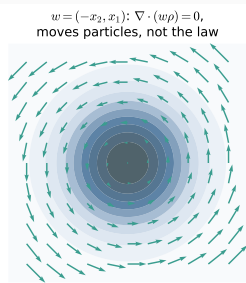
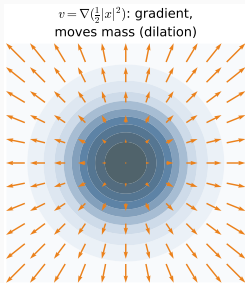
$$dX_t = \left[v_t(X_t) + \frac{\varepsilon}{2} \nabla \log \rho_t(X_t) \right] dt + \sqrt{\varepsilon} dB_t.$$

- ▶ Law of X_t is ρ_t , for every $\varepsilon > 0$. But the joint distribution, e.g. (X_0, X_1) are all different.

Tangent spaces, metric, gradients

The same curve has many velocities

- ▶ If $\nabla \cdot (\rho w) = 0$, then v and $v + w$ produce the same ρ_t .
- ▶ Example: $\rho = N(0, I_2)$ and the rotation $w(x) = (-x_2, x_1)$. Particles spin, the law stays put.
- ▶ Fix it by choosing the velocity with **least kinetic energy** $\mathbb{E} \|v(X)\|^2$.



- ▶ (Definition of Tangent Space)

$$\text{Tan}(\rho) := \overline{\{\nabla g, g \in C_c^\infty\}}^{\text{L}^2(\rho)}.$$

- ▶ The choice of the velocity is not unique, unless

$$v_t \in \text{Tan}(\rho_t)$$

- ▶ This tangent velocity vector will be our notion of gradient of AC curves.

Variational characterization of tangents

- Tangent velocities are projections, hence unique.

$$\text{Tan}(\rho) := \overline{\{\nabla g, g \in C_c^\infty\}}^{L^2(\rho)}$$

- Metric tensor: $v \in \text{Tan}(\rho)$,

$$\|v\|_\rho^2 := \int \|v(x)\|^2 \rho(dx).$$

Examples

- ▶ Heat equation/ heat flow

$$\partial_t \rho_t = \frac{1}{2} \Delta \rho_t.$$

- ▶ Note,

$$\Delta \rho_t = \nabla \cdot (\nabla (\log \rho_t) \rho_t).$$

- ▶ $v_t(x) = -\frac{1}{2} \nabla \log \rho_t(x)$. Gradient type.



$$\|v_t\|_{L^2(\rho_t)}^2 = \frac{1}{4} \int \|\nabla \log \rho_t\|^2 \rho_t dx = \frac{1}{4} \int \frac{\|\nabla \rho_t\|^2}{\rho_t} dx = \frac{1}{4} I(\rho_t).$$

- ▶ Tangent velocity if finite Fisher information. AC curve.

Example

- ▶ Consider OU diffusion:

$$dX_t = -\frac{1}{2}X_t dt + dB_t,$$

B is a standard Brownian motion.

- ▶ ρ_t - Law of X_t
- ▶ Verify tangent velocity

$$v_t(x) = -\frac{x}{2} - \frac{1}{2}\nabla \log \rho_t(x) = -\frac{1}{2}\nabla \left[\frac{\|x\|^2}{2} + \log \rho_t \right].$$

- ▶ $\mathcal{P}_2(\mathbb{R}^d)$ as an infinite dimensional Riemannian manifold
- ▶ AC curves are solutions of cont. eqn.

$$\partial_t \rho_t + \nabla \cdot (v_t \rho_t) = 0.$$

- ▶ Unique tangent velocity

$$v_t \in \overline{\{\nabla g, g \in C_c^\infty\}}^{(\rho_t)}$$

- ▶ Cont. eqn. are marginal laws of a stochastic process where $X_0 \sim \rho_0$ and

$$\dot{X}_t = v_t(X_t).$$

- ▶ $(\rho_t, v_t \ 0 \leq t \leq 1)$ - AC curve, tangent velocity.
- ▶ Length of the curve

$$\text{Length}(\rho) = \int_0^1 \|v_t\|_{L^2(\rho_t)} dt.$$

Theorem

Fix μ, ν . The curve joining the two that minimizes the length is the McCann interpolation. Constant speed geodesic.

Euclidean

Speed $\|\dot{x}_t\|$.

Length $\int_0^1 \|\dot{x}_t\| dt$.

$\|x - y\|^2 = \min \int_0^1 \|\dot{x}_t\|^2 dt$ over paths from x to y .

Geodesic is the straight line segment.

Wasserstein (Otto)

Speed $(\mathbb{E} \|v_t(X_t)\|^2)^{1/2}$, v_t tangent.

Length $\int_0^1 (\mathbb{E} \|v_t(X_t)\|^2)^{1/2} dt$.

$\mathbb{W}_2^2(\mu, \nu) = \min \int_0^1 \mathbb{E} \|v_t(X_t)\|^2 dt$ over curves from μ to ν (Benamou–Brenier).

Geodesic is the McCann interpolation.

Wasserstein gradients

Gradients: the definition

Euclidean

$$\frac{d}{dt}f(x_t) = \nabla f(x_t) \cdot \dot{x}_t.$$

$\nabla f(x)$ is the vector that represents the derivative through the inner product.

Wasserstein (Otto)

$$\frac{d}{dt}F(\rho_t) = \mathbb{E}[\nabla_{\mathbb{W}}F(\rho_t)(X_t) \cdot v_t(X_t)].$$

$\nabla_{\mathbb{W}}F(\rho) \in \text{Tan}(\rho)$ is the vector field that represents the derivative through $\langle \cdot, \cdot \rangle_{\rho}$.

How to compute a Wasserstein gradient

The recipe

1. Write $F(\rho)$ as an expectation in $X \sim \rho$ (and independent copies if needed).
2. Let X move: $\dot{X}_t = v(X_t)$. Differentiate inside the expectation (chain rule), or use $\partial_t \rho = -\nabla \cdot (\rho v)$ and integrate by parts.
3. Rearrange to the form $\mathbb{E}[G(X) \cdot v(X)]$.
4. Read off $\nabla_{\mathbb{W}} F(\rho) = G$. (It will be a gradient.)

Next: five examples. Let's compute together.

Gradient 1: potential energy

► $F(\rho) = \mathbb{E} V(X)$, $X \sim \rho$.

► Chain rule inside $\mathbb{E}$:

$$\frac{d}{dt} \mathbb{E} V(X_t) = \mathbb{E} [\nabla V(X_t) \cdot \dot{X}_t] = \mathbb{E} [\nabla V(X_t) \cdot v_t(X_t)].$$

► Read off: $\boxed{\nabla_{\mathbb{W}} F(\rho) = \nabla V.}$

► $\rho = \delta_x$: $F = V(x)$, gradient $\nabla V(x)$. ✓

Gradient 2: interaction energy

- ▶ $F(\rho) = \frac{1}{2}\mathbb{E} W(X - X')$, X, X' i.i.d. $\sim \rho$, W even.
- ▶ Both particles move:

$$\frac{d}{dt}F = \frac{1}{2}\mathbb{E}[\nabla W(X - X') \cdot (v(X) - v(X'))] = \mathbb{E}[\nabla W(X - X') \cdot v(X)],$$

using symmetry (∇W is odd).

- ▶ Condition on X : $\nabla_{\mathbb{W}} F(\rho)(x) = \mathbb{E}[\nabla W(x - X')] = (\nabla W * \rho)(x).$
- ▶ Example: $W(z) = \frac{1}{2} \|z\|^2$. Then $F = \frac{1}{2}\mathbb{E} \|X - \mathbb{E}X\|^2$ (half the total variance) and

$$\nabla_{\mathbb{W}} F(\rho)(x) = x - \mathbb{E}X.$$

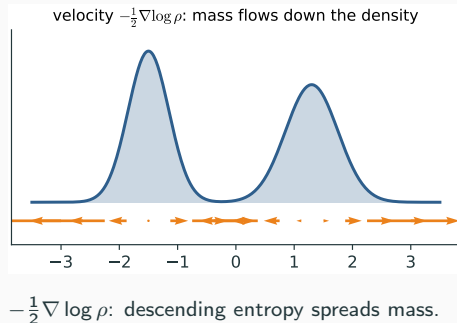
Gradient 3: entropy

- ▶ $\text{Ent}(\rho) = \mathbb{E} \log \rho(X) = \int \rho \log \rho$. Here ρ itself changes, so use densities:

$$\frac{d}{dt} \int \rho_t \log \rho_t = \int \partial_t \rho_t (\log \rho_t + 1)$$

$$= - \int \nabla \cdot (\rho_t v_t) \log \rho_t = \mathbb{E} [\nabla \log \rho_t(X_t) \cdot v_t(X_t)].$$

- ▶ Read off: $\boxed{\nabla_{\mathbb{W}} \text{Ent}(\rho) = \nabla \log \rho}$, the **score**.
- ▶ $\|\nabla_{\mathbb{W}} \text{Ent}(\rho)\|_{\rho}^2 = I(\rho)$, the Fisher information.



Gradient 4: relative entropy

- ▶ $\pi = e^{-V}$ (normalised). $\text{KL}(\rho \mid \pi) = \mathbb{E} \log \frac{\rho}{\pi}(X) = \text{Ent}(\rho) + \mathbb{E} V(X)$.
- ▶ Add Gradients 1 and 3:

$$\nabla_{\mathbb{W}} \text{KL}(\rho \mid \pi) = \nabla \log \rho + \nabla V = \nabla \log \frac{\rho}{\pi},$$

the **relative score**.

- ▶ Its squared length is the **relative Fisher information**:

$$\|\nabla_{\mathbb{W}} \text{KL}(\rho \mid \pi)\|_{\rho}^2 = \mathbb{E} \left\| \nabla \log \frac{\rho}{\pi}(X) \right\|^2 =: I(\rho \mid \pi).$$

- ▶ Only ∇V appears: the normalising constant of π is never needed.

Exercise A

$F(\rho) = \frac{1}{2} \|\mathbb{E}X\|^2$. Compute $\nabla_{\mathbb{W}} F$.

Exercise B

$U_m(\rho) = \frac{1}{m-1} \int \rho^m dx = \frac{1}{m-1} \mathbb{E}[\rho(X)^{m-1}]$, $m > 1$. Compute $\nabla_{\mathbb{W}} U_m$. What happens as $m \downarrow 1$?

Exercise C

Restrict $F(\rho) = \mathbb{E} \frac{X^2}{2}$ to Gaussians $N(m, \sigma^2)$ in $d = 1$. Compute the Euclidean gradient in (m, σ) , and compare with $\nabla_{\mathbb{W}} F = x$.

Gradient flows

Euclidean

$$\dot{x}_t = -\nabla f(x_t).$$

Energy dissipation:

$$\frac{d}{dt} f(x_t) = -\|\nabla f(x_t)\|^2.$$

Wasserstein (Otto)

$$\dot{X}_t = -\nabla_{\mathbb{W}} F(\rho_t)(X_t), \quad \rho_t = \text{Law}(X_t).$$

$$\text{Equivalently } \partial_t \rho_t = \nabla \cdot (\rho_t \nabla \delta F(\rho_t)).$$

$$\frac{d}{dt} F(\rho_t) = -\mathbb{E} \|\nabla_{\mathbb{W}} F(\rho_t)(X_t)\|^2.$$

- ▶ A particle system in which the drift depends on the law of the particles (a McKean–Vlasov ODE).
- ▶ Next: one flow for each gradient we computed.

Jordan, Kinderlehrer & Otto (1998); Otto (2001).

Flow 1 and 2: potential and interaction

- ▶ Potential: $\dot{X}_t = -\nabla V(X_t)$. Every particle runs its own Euclidean gradient descent.
- ▶ Interaction with $W(z) = \frac{1}{2} \|z\|^2$: $\dot{X}_t = -(X_t - \mathbb{E}X_t)$.
 - ▶ $\frac{d}{dt} \mathbb{E}X_t = 0$, so the mean m is fixed and $X_t - m = e^{-t}(X_0 - m)$.
 - ▶ The cloud contracts to its mean, shape preserved. Energy: $F(\rho_t) = e^{-2t}F(\rho_0)$.



Flow 3: the heat equation

1. $F = \frac{1}{2}\text{Ent}$, $\nabla_{\mathbb{W}} F = \frac{1}{2}\nabla \log \rho$. The flow:

$$\dot{X}_t = -\frac{1}{2}\nabla \log \rho_t(X_t), \quad \partial_t \rho_t = \frac{1}{2}\nabla \cdot (\rho_t \nabla \log \rho_t) = \frac{1}{2}\Delta \rho_t.$$

2. The heat equation is the **steepest descent of entropy**.
3. Dissipation: $\frac{d}{dt} \frac{1}{2}\text{Ent}(\rho_t) = -\frac{1}{4}I(\rho_t)$.

Flow 4: Fokker–Planck and Langevin

1. $F = \frac{1}{2}\text{KL}(\cdot | \pi)$, $\pi = e^{-V}$. The flow:

$$\partial_t \rho_t = \frac{1}{2} \Delta \rho_t + \frac{1}{2} \nabla \cdot (\rho_t \nabla V),$$

the Fokker–Planck equation of the **Langevin diffusion**

$$dX_t = -\frac{1}{2} \nabla V(X_t) dt + dW_t.$$

Discretising: explicit versus implicit Euler

Euclidean

Explicit: $x_{k+1} = x_k - \tau \nabla f(x_k)$.

Implicit (proximal):

$$x_{k+1} = \arg \min_y \left[\frac{1}{2\tau} \|y - x_k\|^2 + f(y) \right].$$

First-order condition: $\frac{x_k - x_{k+1}}{\tau} = \nabla f(x_{k+1})$.

Wasserstein (Otto)

Explicit needs $\nabla \log \rho$: undefined for samples.

JKO scheme:

$$\rho_{k+1} = \arg \min_{\mu} \left[\frac{1}{2\tau} \mathbb{W}_2^2(\mu, \rho_k) + F(\mu) \right].$$

First-order condition: $\frac{T - \text{Id}}{\tau} = \nabla_{\mathbb{W}} F(\rho_{k+1})$, T the Brenier map $\rho_{k+1} \rightarrow \rho_k$.

The implicit scheme only uses distances and values of F , so it makes sense in $\mathcal{P}_2$ (De Giorgi's minimising movements).

Jordan, Kinderlehrer & Otto (1998).

JKO by hand: the heat flow on Gaussians

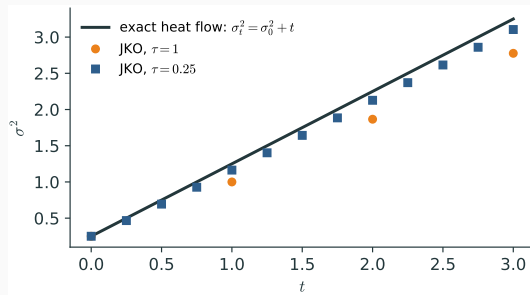
1. $F = \frac{1}{2}\text{Ent}$, $\rho_k = N(m, s_k^2)$. On Gaussians the step is

$$\min_s \frac{1}{2\tau}(s - s_k)^2 - \frac{1}{2} \log s.$$

2. First-order condition: $s^2 - s_k s - \frac{\tau}{2} = 0$,

$$s_{k+1} = \frac{1}{2} \left(s_k + \sqrt{s_k^2 + 2\tau} \right).$$

3. Expand: $s_{k+1} = s_k + \frac{\tau}{2s_k} + O(\tau^2)$, an implicit Euler step for $\dot{\sigma} = \frac{1}{2\sigma}$.



JKO iterates against the exact heat flow $\sigma_t^2 = \sigma_0^2 + t$.

Geodesics and geodesic convexity

Geodesic convexity: second derivatives

Euclidean

f is λ -convex iff along every line

$$\frac{d^2}{dt^2} f(x + tu) \geq \lambda \|u\|^2.$$

$$\nabla^2 f \succeq \lambda I.$$

Wasserstein (Otto)

F is λ -geodesically convex iff along every geodesic

$$\frac{d^2}{dt^2} F(\rho_t) \geq \lambda \mathbb{W}_2^2(\rho_0, \rho_1) = \lambda \mathbb{E} \|U\|^2,$$

where $X_t = X + tU$, $U = T(X) - X$.

- ▶ Recipe: write $F(\rho_t)$ as an expectation in $X + tU$ and differentiate **twice** inside $\mathbb{E}$.
- ▶ Gaussian family: λ -convexity in (m, σ) is ordinary λ -convexity in the plane.

McCann (1997); Ambrosio, Gigli & Savaré (2008) Ch. 9.

The dictionary

	Euclidean	Otto
point; velocity	$x; \dot{x}$	$\rho = \text{Law}(X); v \text{ with } \dot{X} = v(X)$
inner product	$u \cdot v$	$\mathbb{E}[u(X) \cdot v(X)]$
distance	$\ x - y\ $	$\mathbb{W}_2(\mu, \nu)$
geodesic	$x + t(y - x)$	$X + t(T(X) - X)$
gradient	∇f	$\nabla_{\mathbb{W}} F = \nabla \delta F$
examples	$\nabla V, x - y$	$\nabla V, \nabla W * \rho, \nabla \log \rho, \text{Id} - T$
Hessian $\succeq \lambda$	$\nabla^2 f \succeq \lambda$	$\frac{d^2}{dt^2} F(\rho_t) \geq \lambda \mathbb{E} \ U\ ^2$
gradient flow	$\dot{x} = -\nabla f$	heat, Fokker–Planck, porous medium
implicit Euler	proximal step	JKO
PL inequality	$\lambda(f - f^*) \leq \frac{1}{2} \ \nabla f\ ^2$	log-Sobolev

Three lectures in one slide

- ▶ **Geometry.** Brenier maps are gradients of convex functions; McCann interpolations are geodesics; Benamou–Brenier.
- ▶ **Probability.** Entropy regularisation gives Schrödinger bridges, computed by Sinkhorn or iterated Markov projections. Large deviations bring back the geometry.
- ▶ **Calculus.** Otto calculus is Euclidean calculus inside an expectation. Heat flow and Langevin diffusions are gradient flows of entropy and KL; convexity gives rates.

Homework: do it yourself

1. (Porous medium) Write the gradient flow of U_m from Exercise B and show it is $\partial_t \rho = \Delta \rho^m$.
2. (Exponentials) For $\rho_t = \text{Exp}(t)$, $0 < t \leq 1$, find the tangent velocity and the speed. Does the curve have finite length?
3. (Gaussians in $\mathbb{R}^d$) For $X = m + \Sigma^{1/2}Z$, show that the tangent velocity of $t \mapsto \text{Law}(X_t)$ is affine. Compute $\nabla_{\mathbb{W}} \text{Ent}$ and the heat flow in terms of (m, Σ) .
4. (JKO for a potential) Compute one JKO step for $F(\rho) = \mathbb{E} \frac{X^2}{2}$ from $\rho_k = N(m, s^2)$ and compare with $\dot{X} = -X$.
5. (Aggregation) For $W(z) = \frac{1}{4}z^4$ in $d = 1$, write the flow $\dot{X}_t = -\mathbb{E}[(X_t - X'_t)^3 | X_t]$ and compute $\frac{d}{dt} \text{Var}(X_t)$.

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Thank you!